



Project Proposal

Team Members & Project Title

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Electronic Schematic Symbol Classifier					

Topic & Summary

Topic of Interest

We are investigating electronic schematics. The goal is to be able to classify individual components of an electronic schematic, and scale up to parse the entire schematic at once.

Summary: 1. Why Is This Topic Important? 2. How Will Data Be Used? 3. Potential Impact

1. Hand-drawn circuit schematics are common in electronics education and early-stage prototyping, but their legibility varies widely, forcing engineers and students to manually transcribe sketches into digital formats before they can simulate or verify their designs. This process is slow, error-prone, and slows down iteration, especially in classrooms or collaborative settings where schematics are often drawn informally. By applying computer vision and object detection to automatically identify circuit components from hand-drawn sketches, this project addresses a practical bottleneck while also tackling a broader challenge: building reliable models for messy, real-world visual data, an area where well-labeled datasets are typically scarce.

2. We will use a dataset of hand-drawn circuit schematic images, where individual components are labeled. This labeled data will be used to train a CNN-based object detection model, allowing it to learn the visual patterns associated with each component symbol despite variations in handwriting and drawing style.

3. This project could help students quickly verify their hand-drawn circuit designs without manually redrawing them in CAD software, making it easier to catch mistakes early and learn circuit conventions faster. For engineers and educators, it could streamline the process of digitizing informally sketched ideas, whether from a whiteboard brainstorm or a notebook prototype, into structured, shareable formats. More broadly, the project demonstrates how computer vision can be applied to noisy, real-world visual data, offering a foundation that could extend to other domains involving hand-drawn or informally structured technical diagrams, such as flowcharts or architectural sketches.

Machine Learning Algorithm(s)

Computer Vision (CNNs)	Computer Vision (CV)	YOLO (You Only Look Once)
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Research Question

Can we classify essential electronic schematic components given a handwritten image of them?

Dataset(s)

• Name: Digitize-HCD: A Dataset for Digitization of Handwritten Circuit Diagrams • Source: Mendely Data • Link: Digitize-HCD: A Dataset for Digitization of Handwritten Circuit Diagrams - Mendeley Data	• Name: Handwritten Schematics - CGHD • Source: Kaggle • Link: https://www.kaggle.com/datasets/johannesbayer/cghd1152
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Training & Testing Considerations

The primary dataset (Digitize-HCD from Mendeley Data) consists of image-based representations of electronic circuit schematics with bounding box coordinates for individual components. It features 1,277 handwritten circuit diagrams sourced from 150 volunteers, ensuring high structural diversity in handwriting styles and drawing techniques. Given our project timeline and scope, we are training on a targeted taxonomy of approximately 17 foundational electronic components. A secondary dataset includes more examples and components that we may use to expand our model given the time.

Sources of Bias

There are different standards used across the world which result in different symbols indicating the same parts across the world. The dataset might bias one of those standards, and thus not recognize some components.	Because we're looking at handwritten schematics, there are conditions in the data that may influence the model such as lighting, quality of handwriting, and more.	
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Mitigation Strategies

Sample appropriately ensuring fair representation across regional standards, individual components, and other factors.	Implement data augmentation and preprocessing to provide more data and examples to train on, and so the model learns to adapt to different conditions.
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Citations

N. Ahmed, M.F. Adnan, A. Shafiullah, H.J. Parash, Md.S. Rahman, I.C. Akib, G. Sarowar,	Johannes Bayer. (2025). Handwritten Schematics - CGHD [Dataset]. Kaggle. https://doi.org	[Citation]	[Citation]	[Citation]	[Citation]	[Citation]
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“Digitize-HCD : A dataset for digitization of handwritten circuit diagrams,” Data in Brief, vol. 59, p. 111315, Apr. 2025, doi: 10.1016/j.dib.2025.111315	/10.34740/KAGGLE/DSV/13548113					
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